

Pediatric Medications Reference Guide

Common Over-the-Counter and Prescription Medications for Children

This guide covers 10 medications commonly used in pediatric care. Each entry includes: what it is, what it treats, available forms, age restrictions, safety information, side effects, drug interactions, and storage.

Medications: Acetaminophen (Tylenol), Ibuprofen (Motrin/Advil), Amoxicillin, Cetirizine (Zyrtec), Diphenhydramine (Benadryl), Albuterol (ProAir/Ventolin), Prednisolone, Ondansetron (Zofran), Oral Rehydration Solutions (Pedialyte), Hydrocortisone Cream.

■ **IMPORTANT:** This reference does NOT provide specific dosages. Dosing for children is based on weight and must be determined by a healthcare provider or pharmacist. Always follow the dosing instructions on the medication packaging or as directed by your pediatrician.

Sources: FDA DailyMed, MedlinePlus (NIH/NLM), AAP HealthyChildren.org, CDC.

■ **DISCLAIMER:** This document is for informational and educational purposes only. It is not a substitute for professional medical advice.

1. Acetaminophen (Tylenol)

What It Is

Acetaminophen (also called paracetamol in some countries) is an over-the-counter pain reliever and fever reducer. It is one of the most commonly used medications for children. Brand names include Tylenol, FeverAll (suppositories), Tempra, Panadol, and many generic and store brands. It works by affecting the areas of the brain that receive pain signals and regulate body temperature.

What It Treats

- Fever reduction (antipyretic)
- Mild to moderate pain: headaches, toothaches, earaches, sore throat
- Muscle aches and body pain from colds or flu
- Pain after immunizations
- Post-surgical or post-injury pain (mild)

Available Forms for Children

- Liquid oral suspension: 160 mg per 5 mL (standard concentration for all infants' and children's products since 2011)
- Chewable tablets: 160 mg per tablet
- Dissolvable powder packets: 160 mg per packet (ages 6 to 11)
- Rectal suppositories: 80 mg, 120 mg, 325 mg, 650 mg (FeverAll brand)
- Regular strength tablets: 325 mg (for older children)

Age Restrictions and General Dosing Principles

■ Do not give to infants under 12 weeks (3 months) of age without a doctor's direction. Fever in the first 12 weeks of life may signal a serious infection requiring immediate medical evaluation.

- Dosing is based on the child's WEIGHT, not age. Weight-based dosing is more accurate and safer
- Given every 4 to 6 hours as needed
- Do not exceed 5 doses in 24 hours
- Always use the measuring device (syringe or cup) that came with the product. Never use household spoons
- For children under 2 years or under 24 pounds, consult a pediatrician or pharmacist for the correct dose

Important Safety Information

■ Acetaminophen overdose can cause severe, potentially fatal liver damage. This is the leading cause of acute liver failure in the United States.

- Check ALL other medications the child is taking. Acetaminophen is a hidden ingredient in many combination cough, cold, and flu products. Giving multiple products containing acetaminophen can lead to overdose
- Do not use extended-release (650 mg) products in children
- Do not use adult extra-strength (500 mg) products in children under 12

- Store out of reach of children. Child-resistant packaging should be used
- If a child vomits a dose within 20 minutes, it is generally safe to give another dose. If more than 20 minutes have passed, wait until the next scheduled dose

Common Side Effects

When used as directed, acetaminophen has very few side effects. Rare side effects may include nausea, rash, or allergic reaction. Serious allergic reactions (difficulty breathing, swelling of face/throat) are extremely rare but require immediate medical attention.

Drug Interactions

- Warfarin (blood thinner): acetaminophen may increase warfarin's effect
- Other medications containing acetaminophen: risk of unintentional overdose
- Certain seizure medications (carbamazepine, phenobarbital, phenytoin) may increase the risk of liver damage from acetaminophen
- Always inform the doctor or pharmacist of all medications the child is taking

Overdose Warning Signs

■ If you suspect an acetaminophen overdose, call Poison Control at 1-800-222-1222 or 911 immediately, even if the child seems fine. Liver damage may not show symptoms for 24 to 72 hours.

- Early symptoms (first 24 hours): nausea, vomiting, sweating, loss of appetite, general malaise
- Later symptoms (24 to 72 hours): upper right abdominal pain, dark urine, yellowing of skin or eyes (jaundice)
- A child may appear well initially even after a dangerous overdose

Storage

Store at room temperature (68 to 77 degrees Fahrenheit). Keep away from moisture and heat. Do not freeze liquid formulations. Check expiration dates regularly and replace expired products.

Source: FDA DailyMed, MedlinePlus/NIH, AAP HealthyChildren.org, Tylenol Professional Prescribing Info

2. Ibuprofen (Motrin / Advil)

What It Is

Ibuprofen is a nonsteroidal anti-inflammatory drug (NSAID) available over the counter. It reduces fever, relieves pain, and reduces inflammation. Brand names include Motrin, Advil, and many generic brands. Unlike acetaminophen, ibuprofen also has anti-inflammatory properties, making it particularly effective for pain involving swelling or inflammation.

What It Treats

- Fever reduction
- Mild to moderate pain: headaches, toothaches, earaches, sore throat
- Muscle aches and body pain
- Inflammation: sprains, strains, minor injuries
- Menstrual cramps (older children and adolescents)
- Pain from ear infections, dental procedures, or minor surgery

Available Forms for Children

- Liquid oral suspension: 100 mg per 5 mL (infants' concentrated drops and children's suspension)
- Chewable tablets: 100 mg per tablet
- Junior strength tablets or capsules: 100 mg
- Adult tablets: 200 mg (for older children per doctor's guidance)

Age Restrictions and General Dosing Principles

■ Do NOT give ibuprofen to infants younger than 6 months of age.

- Dosing is based on the child's WEIGHT, not age
- Given every 6 to 8 hours as needed
- Do not exceed 3 to 4 doses in 24 hours
- Give with food or milk to reduce stomach irritation
- Use the measuring device that came with the product
- For children under 2 years, consult a pediatrician for the correct dose

Important Safety Information

■ Do NOT give ibuprofen during chickenpox. It may increase the risk of severe skin infections (necrotizing fasciitis).

- Do not give on an empty stomach. Take with food, milk, or a snack to prevent stomach irritation
- Do not give to a child who is dehydrated or vomiting frequently, as it can affect kidney function
- Do not use if the child has a known allergy to ibuprofen, aspirin, or other NSAIDs
- Use with caution in children with asthma, as NSAIDs can trigger bronchospasm in some children
- Do not give aspirin or other NSAIDs at the same time as ibuprofen

- If alternating with acetaminophen (only if directed by a doctor), keep a written log to avoid confusion and overdose

Common Side Effects

- Stomach pain, nausea, or upset stomach (most common, reduced by taking with food)
- Diarrhea or constipation
- Dizziness or headache
- Rash (discontinue and contact doctor)
- Serious but rare: stomach bleeding, kidney problems, allergic reaction

Drug Interactions

- Aspirin and other NSAIDs: increased risk of stomach bleeding
- Blood thinners (warfarin): increased bleeding risk
- Lithium: ibuprofen can increase lithium levels
- Methotrexate: ibuprofen can increase methotrexate toxicity
- ACE inhibitors and diuretics: reduced effectiveness
- Corticosteroids: increased risk of stomach bleeding when combined

When NOT to Give Ibuprofen

- Child is under 6 months old
- Child has chickenpox
- Child is dehydrated or has been vomiting
- Child has kidney disease or a bleeding disorder
- Child is allergic to ibuprofen or aspirin
- Before surgery (may increase bleeding, ask surgeon)

Storage

Store at room temperature (68 to 77 degrees Fahrenheit). Keep tightly closed and away from moisture. Do not freeze.

Source: FDA DailyMed, MedlinePlus/NIH, AAP Ibuprofen Guidelines, Motrin Prescribing Info

3. Amoxicillin

What It Is

Amoxicillin is a prescription antibiotic in the penicillin family. It is the most commonly prescribed antibiotic for children. It works by stopping the growth of bacteria. Amoxicillin is effective only against bacterial infections and does NOT work against viral infections such as the common cold, flu, or most sore throats. Brand names include Amoxil and many generic versions.

What It Treats

- Ear infections (acute otitis media): the most common reason it is prescribed for children
- Strep throat (Group A Streptococcal pharyngitis)
- Sinus infections (bacterial sinusitis)
- Pneumonia (community-acquired, bacterial)
- Urinary tract infections
- Skin infections
- Some dental infections
- H. pylori infection (in combination with other medications)

Available Forms

- Oral suspension (liquid): available in 125 mg/5 mL, 200 mg/5 mL, 250 mg/5 mL, and 400 mg/5 mL concentrations. Must be reconstituted (mixed with water) by the pharmacist
- Chewable tablets: 125 mg, 250 mg
- Capsules: 250 mg, 500 mg
- Tablets: 500 mg, 875 mg

Dosing Principles

- Prescribed by a doctor based on the child's weight and the type of infection
- Typically given 2 or 3 times per day depending on the formulation and infection
- Course length is usually 5 to 10 days depending on the infection. For strep throat, a full 10-day course is standard

■ **Complete the ENTIRE prescribed course even if the child feels better after a few days. Stopping antibiotics early can lead to the infection returning and can contribute to antibiotic resistance.**

- Can be given with or without food. Taking with food may reduce stomach upset
- Shake liquid suspension well before each dose
- Measure liquid with the oral syringe or measuring device provided by the pharmacy

Important Safety Information

- Tell the doctor if the child has ever had an allergic reaction to penicillin, amoxicillin, ampicillin, or any cephalosporin antibiotic
- Allergic reactions can range from mild rash to severe anaphylaxis. A mild rash that appears several days into treatment may be a non-allergic reaction (common with amoxicillin, especially with viral infections like

mono)

- If the child develops hives, difficulty breathing, or swelling of the face, lips, or throat, stop the medication and seek emergency care immediately
- Amoxicillin can reduce the effectiveness of oral contraceptives (relevant for adolescents)
- Inform the doctor of all medications the child is taking

Common Side Effects

- Diarrhea (most common, due to disruption of normal gut bacteria)
- Nausea, vomiting, or stomach pain
- Diaper rash or yeast infection (thrush) due to disruption of normal flora
- Skin rash (non-allergic amoxicillin rash is common, especially with viral infections)
- Contact the doctor if side effects are severe or if a rash with hives or swelling develops

Probiotics During Antibiotic Use

Some pediatricians recommend giving a probiotic (such as Culturelle Kids or Florastor Kids) during and for a week after antibiotic treatment to help restore healthy gut bacteria and reduce diarrhea. Give the probiotic at least 2 hours apart from the antibiotic dose. Ask your pediatrician if this is appropriate.

Storage

- Reconstituted liquid suspension must be refrigerated and discarded after 14 days
- Capsules and tablets stored at room temperature away from moisture and heat
- Return unused or expired antibiotics to a pharmacy for proper disposal. Do not flush or put in household trash

Source: FDA DailyMed Amoxicillin Label, MedlinePlus/NIH, AAP Antibiotic Guidelines

4. Cetirizine (Zyrtec)

What It Is

Cetirizine is a second-generation antihistamine available over the counter. It blocks the effects of histamine, a substance produced by the body during an allergic reaction. It causes less drowsiness than first-generation antihistamines like diphenhydramine. Brand names include Zyrtec and many generic versions. It provides 24-hour allergy relief with a single daily dose.

What It Treats

- Seasonal allergies (hay fever): sneezing, runny nose, itchy and watery eyes, itchy nose or throat
- Year-round (perennial) allergies: dust mites, pet dander, mold
- Hives (urticaria): itchy, raised welts on the skin
- Itching from eczema or other allergic skin conditions (as an adjunct treatment)
- Itching from insect bites

Available Forms for Children

- Liquid oral syrup: 1 mg per mL (Children's Zyrtec)
- Chewable tablets: 5 mg, 10 mg
- Tablets: 10 mg
- Dissolving tablets: 10 mg

Age Restrictions and General Dosing Principles

- Ages 6 months to 23 months: use only under a doctor's direction
- Ages 2 to 5 years: typically a lower dose
- Ages 6 years and older: standard dose
- Given once daily (lasts 24 hours)
- Can be given with or without food
- Can be given in the morning or evening. If drowsiness occurs, give at bedtime

Important Safety Information

- Generally well-tolerated in children with fewer side effects than first-generation antihistamines
- Tell the doctor if the child has kidney or liver disease, as dose adjustment may be needed
- Avoid giving with other antihistamines to prevent excessive drowsiness
- Cetirizine does not treat the underlying cause of allergies. It only manages symptoms
- If hives do not improve or worsen, or if the child has difficulty breathing or swelling of the face/throat, seek emergency care

Common Side Effects

- Drowsiness or sleepiness (less common than with diphenhydramine, but still possible)
- Dry mouth
- Headache

- Stomach pain
- Fatigue
- Rarely: hyperactivity or irritability in young children (paradoxical reaction)

Drug Interactions

- Other antihistamines: increased sedation
- Sedatives, sleep aids, or anxiety medications: increased drowsiness
- Alcohol: increased drowsiness (relevant for adolescents)
- Theophylline: may increase cetirizine blood levels

Storage

Store at room temperature (68 to 77 degrees Fahrenheit). Keep the bottle tightly closed. Protect from moisture. Check expiration dates.

Source: FDA DailyMed Cetirizine Label, MedlinePlus/NIH, AAP Allergy Guidelines

5. Diphenhydramine (Benadryl)

What It Is

Diphenhydramine is a first-generation antihistamine available over the counter. It blocks histamine and also has sedating effects. Brand names include Benadryl, Banophen, and many generic and store brands. It acts quickly (within 15 to 30 minutes) but lasts only 4 to 6 hours and causes significant drowsiness. It is most useful for acute allergic reactions and itching.

What It Treats

- Acute allergic reactions: hives, itching, swelling, runny nose, sneezing, watery eyes
- Itching from insect bites, poison ivy, eczema flares, or chickenpox
- Motion sickness (prevention and treatment)
- As an adjunct treatment during an allergic reaction (not a replacement for epinephrine in anaphylaxis)
- Occasional use as a sleep aid (only under a doctor's direction)

Available Forms for Children

- Liquid oral syrup: 12.5 mg per 5 mL (Children's Benadryl)
- Chewable tablets: 12.5 mg
- Tablets and capsules: 25 mg
- Topical cream, gel, or spray for skin application (used for localized itching)
- Dissolving strips: 25 mg

Age Restrictions and General Dosing Principles

■ Do NOT give diphenhydramine to children under 2 years of age unless directed by a doctor.

- Dosing is based on the child's WEIGHT
- Given every 4 to 6 hours as needed
- Do not exceed 6 doses in 24 hours
- For allergic reactions, oral liquid works fastest. Measure with the provided dosing device
- Topical diphenhydramine (cream/gel) should not be used at the same time as oral diphenhydramine to avoid excessive dosing

Important Safety Information

■ Diphenhydramine overdose in children can cause serious symptoms including hallucinations, seizures, and cardiac arrhythmias. Keep out of reach of children.

- Causes significant drowsiness. Do not use when the child needs to be alert (school, sports)
- Do not use to make children sleepy for travel or convenience. This is not an appropriate use
- Do not use topical diphenhydramine on large areas of the body, broken skin, or blistered skin
- Do not combine oral and topical diphenhydramine products
- Not recommended for daily allergy management in children. Cetirizine or loratadine are better choices for ongoing allergies because they cause less drowsiness and last longer

- In some young children, diphenhydramine can cause a paradoxical reaction: hyperactivity, agitation, and excitability instead of drowsiness

Common Side Effects

- Drowsiness and sedation (most common and expected)
- Dry mouth, nose, and throat
- Dizziness
- Constipation
- Thickening of mucus in the airways
- Blurred vision
- Paradoxical excitability in young children

Drug Interactions

- Other antihistamines (cetirizine, loratadine): do not combine
- Sedatives, sleep aids, or medications that cause drowsiness: excessive sedation
- MAO inhibitors: can intensify and prolong anticholinergic effects
- Topical diphenhydramine with oral diphenhydramine: risk of overdose

When to Use Diphenhydramine vs. Other Antihistamines

Diphenhydramine is best for acute, short-term situations: a sudden allergic reaction, severe itching that needs fast relief, or a single night of itching from a bug bite. For daily allergy management (seasonal or year-round allergies), cetirizine (Zyrtec) or loratadine (Claritin) are preferred because they last 24 hours, cause less drowsiness, and are dosed once daily.

Storage

Store at room temperature (68 to 77 degrees Fahrenheit). Protect from moisture and light. Keep tightly closed.

Source: FDA DailyMed Diphenhydramine Label, MedlinePlus/NIH, AAP Allergy Guidelines

6. Albuterol (ProAir / Ventolin)

What It Is

Albuterol is a prescription bronchodilator (a medication that opens up the airways). It is classified as a short-acting beta-2 agonist (SABA). It is the most commonly used rescue medication for asthma in children. Brand names include ProAir, Ventolin, and Proventil. It works within minutes by relaxing the muscles around the airways, allowing the child to breathe more easily. It is a quick-relief (rescue) medication, not a daily controller medication.

What It Treats

- Asthma attacks (acute bronchospasm): the primary use in children
- Wheezing from bronchiolitis or viral respiratory infections
- Exercise-induced bronchospasm: used 15 to 30 minutes before physical activity to prevent symptoms
- Croup-related wheezing (sometimes, under doctor's direction)
- Chronic obstructive pulmonary disease (COPD) in adults

Available Forms

- Metered-dose inhaler (MDI): 90 mcg per actuation. Used with a spacer device (and face mask for young children)
- Nebulizer solution: 0.63 mg/3 mL, 1.25 mg/3 mL, 2.5 mg/3 mL. The nebulizer turns liquid medication into a mist that the child breathes through a mask or mouthpiece
- Dry powder inhaler: for older children who can generate sufficient inspiratory force

How to Use

- Always use an MDI with a spacer device (a chamber that attaches to the inhaler). This delivers more medication to the lungs and less to the mouth and throat. For children under 4 to 5, use a spacer with a face mask
- Shake the inhaler well before each use. Prime it (spray into the air) if it is new or has not been used for 2 weeks
- For an MDI with spacer: have the child breathe in slowly and deeply, hold breath for 10 seconds (or as long as able), then breathe out. Wait 1 minute between puffs if multiple puffs are prescribed
- For a nebulizer: have the child breathe normally through the mask or mouthpiece for 5 to 10 minutes until the medication is gone
- Rinse the mouth with water after use to prevent throat irritation

General Dosing Information

- For acute asthma symptoms: typically 2 puffs via MDI with spacer, or one nebulizer treatment. May be repeated every 4 to 6 hours as needed
- For exercise prevention: typically 2 puffs 15 to 30 minutes before exercise
- If a child needs albuterol more than 2 days per week (not counting pre-exercise use), the asthma may not be well-controlled and the child's asthma action plan should be reviewed with the doctor

■ If albuterol does not improve symptoms or symptoms return within 4 hours, seek medical care. This may indicate a severe asthma attack requiring emergency treatment.

Common Side Effects

- Tremor or shakiness (most common)
- Increased heart rate (tachycardia)
- Nervousness, restlessness, or hyperactivity
- Headache
- Throat irritation or cough (reduced by using a spacer and rinsing mouth)
- These side effects are usually mild and resolve quickly

Important Safety Information

- Albuterol is a rescue medication for acute symptoms. It does NOT treat the underlying inflammation of asthma
- Children with asthma also need a daily controller medication (inhaled corticosteroid) if symptoms are frequent
- Check the inhaler's dose counter regularly and replace before it runs out
- Keep a rescue inhaler with the child at school, at home, and during sports or physical activities
- Inform the school nurse and provide a copy of the child's asthma action plan

Storage

Store the MDI at room temperature (59 to 77 degrees Fahrenheit). Do not puncture or expose to high heat. Keep away from open flame. Protect from freezing. Nebulizer solution should be stored as directed on the package.

Source: FDA DailyMed Albuterol Label, MedlinePlus/NIH, AAP Asthma Management Guidelines

7. Prednisolone / Prednisone (Oral Corticosteroid)

What It Is

Prednisolone (brand names: Prelone, Orapred) and prednisone are prescription oral corticosteroids (steroids). They are powerful anti-inflammatory medications that reduce swelling, redness, and immune system activity. Prednisolone is the liquid form most commonly used in young children. Prednisone is available in tablets for older children. These are NOT anabolic steroids used by athletes.

What It Treats in Children

- Croup: a short course (1 to 3 days) dramatically reduces airway swelling. Often a single dose is sufficient for mild to moderate croup
- Asthma exacerbations (flare-ups): a 3 to 5 day course reduces airway inflammation during an asthma attack
- Severe allergic reactions: reduces swelling and inflammation
- Inflammatory conditions: certain autoimmune conditions, severe eczema flares, nephrotic syndrome
- Poison ivy or poison oak: severe cases with widespread rash

Available Forms

- Prednisolone oral solution: 15 mg per 5 mL (most common for young children)
- Prednisolone ODT (orally disintegrating tablets): 10 mg, 15 mg, 30 mg
- Prednisone tablets: 1 mg, 2.5 mg, 5 mg, 10 mg, 20 mg, 50 mg
- The liquid has a bitter taste. Mixing with juice, chocolate syrup, or giving with a popsicle can help mask the taste

Dosing Principles

- Prescribed by a doctor. Dose depends on the child's weight and the condition being treated
- Short courses (burst therapy): typically 3 to 5 days for asthma or croup. A short course usually does not require tapering (gradually reducing the dose)
- Longer courses (more than 7 to 14 days): may require gradual tapering to allow the body's own cortisol production to resume. Never stop a long course abruptly
- Usually given once daily in the morning, or split into two doses
- Give with food to reduce stomach irritation

Important Safety Information

■ Do not stop a prolonged course of steroids abruptly. The body suppresses its own cortisol production during steroid use, and sudden withdrawal can cause adrenal crisis.

- Short courses (3 to 5 days) are generally very safe with minimal lasting side effects
- Steroids suppress the immune system. During treatment, the child may be more susceptible to infections
- Avoid exposure to chickenpox or measles during steroid treatment if the child has not been vaccinated or has not had these diseases
- Inform the doctor if the child is receiving any vaccines, as live vaccines should generally not be given during steroid treatment

- Monitor for mood or behavior changes (irritability, hyperactivity, mood swings), which are common during short courses and resolve after stopping

Common Side Effects (Short Course)

- Increased appetite and weight gain
- Mood changes: irritability, hyperactivity, difficulty sleeping, mood swings
- Upset stomach (take with food)
- Increased blood sugar (relevant for diabetic children)
- Facial flushing
- These side effects are temporary and resolve when the medication is stopped

Storage

Prednisolone solution should be stored at room temperature or refrigerated depending on the brand. Check the label. Prednisone tablets are stored at room temperature away from moisture.

Source: FDA DailyMed Prednisolone Label, MedlinePlus/NIH, AAP Asthma and Croup Guidelines

8. Ondansetron (Zofran)

What It Is

Ondansetron is a prescription anti-nausea medication (antiemetic). It works by blocking serotonin receptors in the brain that trigger nausea and vomiting. Brand name is Zofran. It was originally developed for chemotherapy-induced nausea but is now widely used in pediatric emergency departments and by pediatricians for vomiting from gastroenteritis and other causes.

What It Treats

- Vomiting from gastroenteritis (stomach flu): the most common pediatric use
- Nausea and vomiting that prevents a child from keeping down fluids (risk of dehydration)
- Post-operative nausea and vomiting
- Chemotherapy-induced nausea and vomiting

Available Forms

- Orally disintegrating tablets (ODT): 4 mg, 8 mg. Dissolve on the tongue without water, making them ideal for a vomiting child who cannot swallow a pill
- Regular tablets: 4 mg, 8 mg
- Oral solution: 4 mg per 5 mL
- Injectable form (used in hospital settings)

Dosing Principles

- Prescribed by a doctor. Dose based on child's weight and age
- For vomiting from gastroenteritis, a single dose is often sufficient to stop vomiting long enough for the child to begin oral rehydration
- May be given every 8 hours as needed, typically for a short duration (1 to 2 days)
- The orally disintegrating tablet is placed on the tongue and allowed to dissolve completely before swallowing. No water needed
- Can be given with or without food

Important Safety Information

- Ondansetron does not treat the underlying cause of vomiting. It is used to control the symptom while addressing the root cause (rehydration for gastroenteritis)
- Ondansetron can cause a slight increase in diarrhea in children with gastroenteritis. This is expected because fluids that would have been vomited are instead passing through the GI tract
- Tell the doctor if the child has a heart condition, as ondansetron can affect heart rhythm (QT prolongation) in rare cases
- Inform the doctor of all other medications the child is taking

Common Side Effects

- Headache (most common)
- Constipation or diarrhea

- Dizziness
- Drowsiness
- These are generally mild and temporary

Storage

Store tablets and ODT at room temperature (68 to 77 degrees Fahrenheit). Oral solution may be stored at room temperature. Keep ODT in original packaging until ready to use, as moisture can affect them.

Source: FDA DailyMed Ondansetron Label, MedlinePlus/NIH, AAP Gastroenteritis Management

9. Oral Rehydration Solutions (Pedialyte)

What It Is

Oral rehydration solutions (ORS) are specially formulated drinks that contain the precise balance of water, sugars, and electrolytes (sodium, potassium, chloride) needed to replace fluids lost through vomiting and diarrhea. Pedialyte is the most recognized brand in the United States. Other brands include Enfalyte, DripDrop, and generic store brands. The World Health Organization (WHO) also produces an ORS formula used worldwide. ORS is available without a prescription.

What It Treats

- Dehydration prevention and treatment from gastroenteritis (stomach flu)
- Dehydration from vomiting and/or diarrhea of any cause
- Fluid replacement during illness with fever and poor oral intake
- Mild to moderate dehydration (severe dehydration requires IV fluids in a medical setting)

Available Forms

- Ready-to-drink liquid: available in multiple flavors
- Freezer pops: small frozen popsicles (helpful for children who refuse to drink)
- Powder packets: mix with water according to package directions
- Jello-like form

All forms contain the same essential electrolyte balance. Choose whichever form the child will accept.

How to Give ORS

- For a vomiting child: start with very small amounts. Give 1 teaspoon (5 mL) every 1 to 2 minutes using a syringe, spoon, or small cup
- If the child keeps this down for 15 to 20 minutes, gradually increase to 1 tablespoon (15 mL) every 5 minutes
- Continue increasing as tolerated. The goal is small, frequent sips rather than large amounts that may trigger more vomiting
- For diarrhea without vomiting: offer ORS freely after each loose stool
- For breastfed babies: continue breastfeeding and offer ORS between feeds
- For formula-fed babies: offer ORS and continue formula as tolerated
- Freezer pops are excellent for children who refuse to drink liquid ORS

Important Information

■ Do NOT substitute sports drinks (Gatorade, Powerade), fruit juice, soda, or broth for ORS. These have incorrect electrolyte and sugar ratios and can worsen diarrhea.

- Do not dilute Pedialyte with water or mix with other liquids, as this changes the electrolyte balance
- Once opened, refrigerate and discard unused liquid after 48 hours
- ORS does not stop vomiting or diarrhea. It replaces the fluids and electrolytes being lost

- If the child cannot keep any fluids down for 4 to 6 hours, or shows signs of dehydration (no tears, dry mouth, no wet diapers for 6 to 8 hours, sunken eyes, lethargy), seek medical attention

Storage

Unopened ORS is stored at room temperature. After opening, refrigerate and use within 48 hours. Powder packets are stored at room temperature until mixed. Freezer pops are stored in the freezer.

Source: WHO ORS Guidelines, MedlinePlus/NIH, AAP Gastroenteritis and Rehydration Guidelines

10. Hydrocortisone Cream (Topical Corticosteroid)

What It Is

Hydrocortisone cream is a mild topical corticosteroid available over the counter in 0.5 percent and 1 percent strengths. It reduces inflammation, itching, and redness on the skin. It is the mildest steroid cream available and is generally safe for short-term use in children. Brand names include Cortizone-10, Aveeno Hydrocortisone, and many generic brands. Stronger prescription topical steroids (such as triamcinolone, betamethasone, or clobetasol) are available for more severe conditions but require a doctor's prescription.

What It Treats

- Eczema (atopic dermatitis): mild flare-ups
- Insect bites: reduces itching and swelling
- Contact dermatitis: poison ivy, poison oak, poison sumac
- Minor skin irritations and rashes
- Allergic skin reactions
- Diaper rash (only if recommended by a doctor, not for routine use)
- Mild psoriasis

How to Use

- Apply a thin layer to the affected area only. Do not apply to large areas of the body
- Gently rub in until absorbed
- Apply 1 to 4 times daily depending on the product label or doctor's direction
- For eczema: apply to red, itchy areas, then apply a thick moisturizer over the top to seal in both the medication and moisture
- Do not cover the area with bandages or tight dressings unless directed by a doctor (occlusion increases absorption and potency)
- Wash hands after application (unless the hands are the treated area)

Areas to Avoid

■ Do not apply hydrocortisone to the face, groin, or armpits without a doctor's direction. These areas have thin, sensitive skin that absorbs steroids more readily, increasing the risk of skin thinning and other side effects.

- Do not apply to broken, infected, or weeping skin
- Do not apply near the eyes
- Do not use on diaper rash unless directed by a doctor
- Do not use under tight diapers or plastic pants as this increases absorption

Duration of Use

- Over-the-counter hydrocortisone: do not use for more than 7 consecutive days without consulting a doctor
- Prescription topical steroids: follow the doctor's specific instructions on duration
- If the condition has not improved after 7 days of OTC hydrocortisone, see a doctor for evaluation and possible prescription treatment

- For eczema: use on flare-ups only, not continuously as maintenance. Daily moisturizing is the maintenance treatment

Important Safety Information

- Topical steroids can cause skin thinning (atrophy) if used for too long or on sensitive areas
- Other potential effects from prolonged or excessive use: stretch marks, visible blood vessels, lightening of skin color at the application site
- Do not use to treat fungal infections (ringworm, athlete's foot) as steroids can worsen fungal infections
- If a rash worsens, spreads, or shows signs of infection (increasing redness, warmth, swelling, pus), stop the cream and see a doctor
- Children absorb more topical steroids through their skin than adults, so use the minimum amount needed

Storage

Store at room temperature (68 to 77 degrees Fahrenheit). Keep the tube tightly closed. Do not freeze.

Source: FDA DailyMed Hydrocortisone Label, MedlinePlus/NIH, AAP Eczema Management, National Eczema Association